

Lesson 16: Laplace Transforms - Convolutions & LTI Systems

Professor Dave Explains Differential Equations Master Series · Lesson 16

Video: Laplace Transforms Part 2: Convolutions and LTI Systems **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that in signal processing and control theory, multiplying two Laplace transforms in the frequency domain corresponds to a fascinating operation in the time domain called Convolution: $(f * g)(t)$. Convolution represents a 'sliding weighted memory'—how a physical system smudges or filters an incoming signal over time. Furthermore, the Dirac delta function $\delta(t)$ models sharp hammer blows or lightning strikes, allowing us to find the fundamental impulse response of any Linear Time-Invariant (LTI) system.

2. Core Theoretical Framework & Rigorous Formulations

1. The Convolution Theorem

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s), \quad (f * g)(t) = \int_0^t f(\tau)g(t - \tau)d\tau$$

Inverse property: $\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(\tau)g(t - \tau)d\tau$.

2. Impulses, Switches & Transfer Functions

- **Dirac Delta Function:** $\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}$. Models an instantaneous unit impulse.
- **Heaviside Step Function:** $\mathcal{L}\{\mathcal{U}(t - a)f(t - a)\} = e^{-as}F(s)$. Models switches turned on at $t = a$.
- **Transfer Function $H(s)$:** For an LTI system with zero initial conditions, $Y(s) = H(s)X(s) \implies y(t) = (h * x)(t)$, where $h(t) = \mathcal{L}^{-1}\{H(s)\}$ is the system's impulse response.

3. Fully Solved Whiteboard Problems

Exam Problem: Solving an IVP with a Sudden Impulse Spike

Problem Statement: Solve the oscillator $y'' + 9y = 4\delta(t - \pi)$ with $y(0) = 0, y'(0) = 0$.

Step 1: Apply Laplace Transform with Impulse

$$(s^2 + 9)Y(s) = 4\mathcal{L}\{\delta(t - \pi)\} = 4e^{-\pi s}.$$

Step 2: Isolate System Transfer Response

$$Y(s) = \frac{4e^{-\pi s}}{s^2 + 9} = \frac{4}{3}e^{-\pi s} \left(\frac{3}{s^2 + 9} \right).$$

Step 3: Identify Base Transform and Time Shift

Base function: $\mathcal{L}^{-1}\left\{\frac{3}{s^2 + 9}\right\} = \sin(3t)$. Using the Second Translation Theorem: $\mathcal{L}^{-1}\{e^{-as}F(s)\} = \mathcal{U}(t - a)f(t - a)$ with $a = \pi$.

Step 4: Compute Time-Shifted Sine

$$y(t) = \frac{4}{3}\mathcal{U}(t - \pi)\sin(3(t - \pi)) = \frac{4}{3}\mathcal{U}(t - \pi)\sin(3t - 3\pi).$$

Step 5: Simplify Trigonometric Phase

Recall $\sin(\theta - 3\pi) = -\sin(\theta)$. Therefore: $y(t) = -\frac{4}{3}\mathcal{U}(t - \pi)\sin(3t)$. Before $t = \pi$, the system is completely at rest; at $t = \pi$, the hammer strikes, triggering perpetual harmonic oscillation!

⚠ Common Exam Pitfall & Trapdoor Avoidance

The shift theorem requires the argument of the function to be shifted by the EXACT same delay: $\mathcal{U}(t - a)f(t - a)$. Do not evaluate $\mathcal{U}(t - a)f(t)$ without shifting $t \rightarrow t - a$!

4. Key Takeaways & Exam Strategy Formula Card

🔑 Core Takeaway & Exam Rule

Convolution is frequency multiplication. $\delta(t - t_0)$ delivers an instantaneous kick creating response $h(t - t_0)$.

- Video reference: [Laplace Transforms Part 2: Convolutions and LTI Systems](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.